



KEALEE

design, build, deliver

PROJECT
42525 CARNFORTH CT. ASHBURN, VA. 20148

ADDRESS
42525 CARNFORTH CT, ASHBURN, VA, 20148

JURISDICTION
Loudoun County, Virginia

LOW
ZON
D1

R1
SHEET

C-100 — Existing Conditions and Boundary Plan

SCALE
1" = 30'

STATUS
PRELIM

SHEET NO.

1 OF 5

COORDINATES

EPSG:2248
H: NAD83

V: NAVD88 (feet)

PROFESSIONAL RESPONSIBILITY

Virginia Licensed Surveyor

certifies: boundary_determination, topographic_survey

SEAL AND SIGNATURE

All content on this sheet falls within one professional scope

PLAN TYPE AND PREPARATION

PLAN TYPE	SITE DEVELOPMENT / FINE GRADING
SHEET	C-100 — Existing Conditions and Boundaries
DISCIPLINE	Virginia Licensed Surveyor
DRAWN BY	KEALEE SITE-PLAN ENGINE
DESIGNED BY	
CHECKED BY	

REVISIONS

NO.	DATE

SITE DATA

ZONE	R1	
TAX / PARCEL ID	15436367100	
GROSS TRACT AREA	20,274 SF	0.465 AC
LOT FRONTAGE	—	79.99'
SETBACKS	REQUIRED	PROVIDED
FRONT YARD	35'	35'
SIDE YARD	12'	12'
REAR YARD	35'	35'
COVERAGE	MAX	PROPOSED
LOT COVERAGE	25%	25.0%
BUILDING FOOTPRINT	—	5,069 SF
DATUM		
HORIZONTAL	EPSG:2248	NAD83
VERTICAL	NAVD88 (feet)	

SITE ANALYSIS

1. Gross tract area	20,274 SF
2. Area of dwelling	5,069 SF
3. Wooded area	NOT ESTABLISHED
4. Floodplain area	NOT ESTABLISHED
5. Net lot area	20,274 SF
6. TOTAL AREA DISTURBED	10,652 SF (AT LEAST)
7. SWM REVIEW	CONFIRM WITH LOUDOUN COUNTY

BUILDING DATA

	G	B	FF	SF	HT	STY
42525 CARNFORTH	—	—	—	—	—	—
CT, ASHBURN, VA	—					
20148	—					

— = Not measured - FF finished floor - SF subfloor - HT height - STY storeys. A DASH IS NOT ZERO: the elevation has not been established and must be set from a field-run survey before construction.

DRAINAGE AREA COMPUTATIONS

CATCHMENT	SF	ACRES	C
Total ~ 28.4% impervious	20,274	0.465	
PRE-DEVELOPMENT	SF		
Periowood ~ lawn / meadow	20,274		0.250
Composite C			0.250
POST-DEVELOPMENT	SF		
Proposed roof	5,069		0.95
Driveway and on-lot paving	698		0.16
Periowood ~ lawn	14,508		0.25
Composite C			0.449
Flow path			25% T
Flow path slope			10 ± %
Time of concentration (Kirpich)			5.0 min
Rainfall intensity, 30-yr			6.33 in/hr
Peak discharge PRE: Q = CIA			0.75 cfs
Peak discharge POST: Q = CIA			1.38 cfs
INCREASE TO BE MANAGED			0.66 cfs
Rv = 0.05 ~ 0.0501			0.3560
Water quality volume WQv			517 cft

Rainfall intensity by Kirpich over a 235 ft flow path at 10.2% tan, taken from the 2 ft contours across the catchment. A 5-minute flow path.
 Time of concentration 6.53 min/hr for the 10-year storm at a 5.0 minute duration. NOAA Atlas 14 Vol 2 Ver. 3. PFD5 point estimates, 93.0386 N 77.5302 W.
 Water quality volume on a 1.0 in rainfall. Computed by the Maryland Stormwater Design Manual method for reference only; LOUDOUN COUNTY's own stormwater method governs this lot.

Runoff coefficients: roof and pavement 0.95, average lawn 0.25. Confirm the lawn coefficient against the soil hydrologic group before relying on the composite.
 Sidewalk, curb and gutter and the driveway apron lie in the public right-of-way and drain to the street. They are excluded from the lot catchment and included in the disturbed area.

SEQUENCE OF CONSTRUCTION

1. Pre-construction meeting
2. Obtain necessary permits
3. Notify Miss Utility at 811 at least 48 hours prior to any excavation
4. Install stabilized construction entrance and perimeter sediment control
5. Rough grade the lot
6. Construct dwelling
7. Install water, sewer and storm services and the driveway
8. Fine grade and stabilize all disturbed areas
9. Remove sediment control once the inspector gives written permission

TOTAL ESTIMATED TIME OF CONSTRUCTION: 8 MONTHS

GENERAL NOTES

1. CONTRACTOR SHALL CONTACT MISS UTILITY AT 811 A MINIMUM OF 48 HOURS PRIOR TO ANY EXCAVATION; FIELD-VERIFIED LOCATION AND DEPTH SHALL BE SHOWN.
2. EXISTING UTILITIES ARE SHOWN FROM PUBLIC RECORD AND ARE APPROXIMATE. LOCATION AND DEPTH SHALL BE FIELD-VERIFIED BY TEST PIT.
3. PROPOSED GRADE SHOWN SOLID; EXISTING GRADE SHOWN DASHED.
4. ROUGH EARTHWORK GRADES AND UTILITY ELEVATIONS SHOWN TO TENTH OF A FOOT.
5. BOUNDARY SHOWN IS COMPLIED JURISDICTION GIS, NOT A BOUNDARY SURVEY.
6. COVERED EXISTING PAVEMENT, CURBS AND GUTTER, DRIVEWAY, AND SIDEWALK IN LINE AND GRADE.
7. NO DRIVEWAY CULVERT IS REQUIRED; THIS FRONTAGE IS AN EXISTING CURB AND GUTTER SECTION; NOT AN OPEN DITCH SECTION. DRIVEWAY SHALL BE CARRIED THROUGH CHAIN ENTRANCE BY THE DEPRESSION CURB AT THE APRON PER LOCAL COUNTY STANDARD.
8. DRIVEWAY SHALL BE CARRIED THROUGH REAR DRIVEWAY APRON, REAR DRIVEWAY EASEMENT AND DRIVEWAY TO DRIVEWAY OR APRON.

PROFESSIONAL CERTIFICATION

I HEREBY CERTIFY THAT THESE DOCUMENTS WERE PREPARED OR APPROVED BY ME, AND THAT I AM A DULY LICENSED PROFESSIONAL ENGINEER UNDER THE LAWS OF THE COMMONWEALTH OF VIRGINIA.

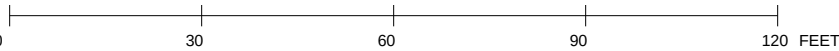
SEAL	SIGNATURE _____	DATE _____
	LICENSE NO. _____	EXPIRATION DATE _____

REVISIONS

SOILS TABLE		USDA NRCS SSBURGO			
MUP UNIT	MUP UNIT NAME	K-FACT	HYDRIC	HYD. GRP	DRAINAGE CLASS
101	Pits, quarry	—	No	—	—
102	Landfills	—	No	—	—
103	Underfills	—	No	—	—
108	Mongle silt loam, 0 to 7 percent slopes, frequently flooded	.32	No	C/D	Somewhat poorly drained
126	Rehobothville cobbly silt loam, 0 to 7 percent slopes, 43% to 59% clay	.32	No	B/D	Somewhat poorly drained
138	Mongle silt loam, 2 to 7 percent slopes	.37	No	B	Well drained
146	Manassas silt loam, 2 to 7 percent slopes	.37	No	B	Medium to well drained
178	Midgessing silt loam, 2 to 7 percent slopes	.37	No	A	Well drained

SOURCE AND ACCURACY
Parcel boundary — jurisdiction gis — Loudoun County — parcel boundaries — AVONMORE, Plat 2023-0020, L...
Existing contours — U.S. Geological Survey — 3D Elevation Program — 3DEP bare-earth DEM —...

GRAPHIC SCALE — 1" = 30'





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Design, build, deliver

PROJECT
42525 CARNFORTH CT, ASHBURN, VA, 20148
ADDRESS
42525 CARNFORTH CT, ASHBURN, VA, 20148
JURISDICTION
Loudoun County, Virginia
ZONE
R1
SHEET
C-200 — Overall Site and Zoning Plan
SCALE
1" = 30'
STATUS
PRELIMINARY
SHEET NO.
2 OF 5
COORDINATES
EPSG-2248
H: NAD83
V: NAVD88 (feet)

PROFESSIONAL RESPONSIBILITY
Virginia Professional Engineer
certifies: site_layout

SEAL AND SIGNATURE

Architect
certifies: architectural_footprint

SEAL AND SIGNATURE

This sheet carries content from more than one professional discipline. Each seal below applies only to the subjects listed beside it. No professional certifies content outside their stated scope, and the sheet is not divided into separate drawings for that purpose.

PLAN TYPE AND PREPARATION

PLANTYPE SITE DEVELOPMENT / FINE GRADING
SHEET C-200 — Overall Site and Zoning Plan
DISCIPLINE Virginia Professional Engineer
DRAWN BY KEALEE SITE-PLAN ENGINE
DESIGNED BY
CHECKED BY

REVISIONS

NO.	DATE	DESCRIPTION

SITE DATA

ZONE	R1
TAX / PARCEL ID	154363671000
GROSS TRACT AREA	20,274 SF 0.465 AC
LOT FRONTAGE	79.99'
SETBACKS	REQUIRED PROVIDED
FRONT YARD	35' 35'
SIDE YARD	12' 12'
REAR YARD	35' 35'
COVERAGE	MAX PROPOSED
LOT COVERAGE	25% 25.0%
BUILDING FOOTPRINT	5,069 SF
DATUM	
HORIZONTAL	EPSG-2248 NAD83
VERTICAL	NAVD88 (feet)

SITE ANALYSIS

1. Gross tract area	20,274 SF
2. Area of dwelling	5,069 SF
3. Wooded area	NOT ESTABLISHED
4. Floodplain area	NOT ESTABLISHED
5. Net lot area	20,274 SF
6. TOTAL AREA DISTURBED	10,652 SF (AT LEAST)
7. SWM REVIEW	CONFIRM WITH LOUDOUN COUNTY

BUILDING DATA

	G	B	FF	SF	HT	STY
42525 CARNFORTH CT, ASHBURN, VA, 20148	—	—	—	—	—	—
Notes: G = ground floor; B = basement; FF = finished floor; SF = square feet; HT = height; STY = storeys. A DASH IS NOT ZERO; the elevation has not been established and must be set from a field-run survey before construction.						

DRAINAGE AREA COMPUTATIONS

CATCHMENT	SF	ACRES	C
Total — 28.4% impervious	20,274	0.465	
PRE-DEVELOPMENT	SF		C
Pervious — lawn / meadow	20,274		0.25
Composite C			0.250
POST-DEVELOPMENT	SF		C
Proposed roof	5,069		0.95
Driveway and on-lot paving	698		0.95
Pervious — lawn	14,508		0.25
Composite C			0.449
Flow path			235 ft
Flow path slope			10.2%
Time of concentration (Kirpich)			5.0 min
Rainfall intensity, 10-yr			6.53 in/hr
Peak discharge PRE, Q = CIA			0.76 cfs
Peak discharge POST, Q = CIA			1.36 cfs
INCREASE TO BE MANAGED			0.60 cfs
Rv = 0.05 + 0.0081			0.390
Water quality volume WQv			517 cf

Time of concentration by Kirpich over a 235 ft flow path at 10.2%, taken from the 2 ft contours across the catchment. A 5-minute flow applies.
Rainfall intensity 6.53 in/hr for the 10-year storm at a 5.0 minute duration. NOAA Atlas 14 Vol. 2 Ver. 3. PFDS point estimate, 39.0386 N 77.5302 W.
Water quality volume on a 1.0 in rainfall. Computed by the Maryland Stormwater Design Manual method for reference only. LOUDOUN COUNTY's own stormwater method governs this lot.
Runoff coefficients: roof and pavement 0.95, average lawn 0.25. Confirm the lawn coefficient against the soil hydrologic group before relying on the composite.
Sidewalk, curb and gutter and the driveway apron lie in the public right-of-way and drain to the street. They are excluded from the lot catchment and included in the disturbed area.

SEQUENCE OF CONSTRUCTION

1. Pre-construction meeting	1 DAY
2. Obtain necessary permits	2 DAYS
3. Notify Miss Utility at 811 at least 48 hours prior to any excavation	1 DAY
4. Install stabilized construction entrance and perimeter sediment control	2 DAYS
5. Rough grade the lot	5 DAYS
6. Construct dwelling	6 MONTHS
7. Install water, sewer and storm services and the driveway	10 DAYS
8. Fine grade and stabilize all disturbed areas	5 DAYS
9. Remove sediment control once the inspector gives written permission	1 DAY

TOTAL ESTIMATED TIME OF CONSTRUCTION: 8 MONTHS

LEGEND

— — —	PROPERTY BOUNDARY
— — —	BRL — BUILDING RESTRICTION LINE
— — —	PROPOSED STRUCTURE
— — —	EXISTING CONTOUR — 2 FT INTERVAL, NAVD88
— — —	PROPOSED CONTOUR — 2 FT INTERVAL
— — —	STREET CENTRELINE
— — —	PROPOSED PAVEMENT — DRIVEWAY / WALK
— — —	EASEMENT
— — —	SANITARY LATERAL (SHC) — HEAVIEST
— — —	WATER SERVICE (WHC)
— — —	STORM DRAIN — TRUNK, RCP
— — —	STORM — HOUSE CONNECTION TO TRUNK
— — —	DRAINAGE SWALE — INVERT AND GRADED SECTION
— — —	RETAINING WALL — TICKS ON THE RETAINED SIDE
— — —	LIMIT OF GROUND COVER — TICKS ON THE STEEP SIDE
— — —	ESD — STORMWATER PRACTICE

PROFESSIONAL CERTIFICATION

I HEREBY CERTIFY THAT THESE DOCUMENTS WERE PREPARED OR APPROVED BY ME, AND THAT I AM A DULY LICENSED PROFESSIONAL ENGINEER UNDER THE LAWS OF THE COMMONWEALTH OF VIRGINIA.

SIGNATURE

DATE

LICENSE NO.

EXPIRATION DATE

REVISIONS

— NONE —

SOILS TABLE

USDA NRCS SSURGO

MAP UNIT	MAP UNIT NAME	K-FACT	HYDRIC	HYD. GRP	DRAINAGE CLASS
101	Pes. quality	—	No	—	—
102	Landfills	—	No	—	—
103	Udorthents	—	—	—	—
108	Mungie silt loam, 0 to 7 percent slopes, frequently flooded	32	No	B/D	Somewhat poorly drained
128	Rohrersville cobbly silt loam, 0 to 7 percent slopes,	43	No	C/D	Somewhat poorly drained
138	SECURITYVILLE silt loam, 0 to 7 percent slopes	37	No	B	Well drained
148	Manassas silt loam, 2 to 7 percent slopes	37	No	B	Moderately well drained
178	Middleburg silt loam, 2 to 7 percent slopes	32	No	A	Well drained

SOURCE AND ACCURACY
Parcel boundary — Jurisdiction gis — Loudoun County — parcel boundaries — AVONMORE, Plat 2023-0020, L...
Existing contours — U.S. Geological Survey — 3D Elevation Program — 3DEP bare-earth DEM —...

GRAPHIC SCALE — 1" = 30'

0 30 60 90 120 FEET



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42525 CARNFORTH CT, ASHBURN, VA, 20148

ADDRESS
42525 CARNFORTH CT, ASHBURN, VA, 20148

JURISDICTION
Loudoun County, Virginia

ZONE
R1

SHEET
C-400 — Grading and Drainage Plan

SCALE
1" = 30'

STATUS
PRELIM

SHEET NO.
3 OF 5

COORDINATE
EPSG:2241

H: NAD83
V: NAVD88 (feet)

PROFESSIONAL R

certifies: grading_design, sedi

SEAL AND SIGNATURE

All content on this sheet falls within one professional scope

PLAN TYPE AND PREPARATION

PLAN TYPE	SITE DEVELOPMENT / FINE GRADING
SHEET	C-400 — Grading and Drainage Plan
DISCIPLINE	Virginia Professional Engineer
DRAWN BY	KEALEE SITE-PLAN ENGINE
DESIGNED BY	
CHECKED BY	

REVISIONS

NO.	DATE

SITE DATA

ZONE	R1	
PLAN / PARCEL ID	154363671000	
GROSS TRACT AREA	20,274 SF	0.465 AC
LOT FRONTAGE	—	79.99'
SETBACKS	REQUIRED	PROVIDED
FRONT YARD	35'	35'
SIDE YARD	12'	12'
REAR YARD	35'	35'
COVERAGE	MAX	PROPOSED
LOT COVERAGE	25%	25.06%
BUILDING FOOTPRINT	—	5,069 SF
DATUM		
HORIZONTAL	EPSG:2248	NAD83
VERTICAL	NAVD88 (feet)	

SITE ANALYSIS

1. Gross tract area	20,274 SF
2. Area of dwelling	5,069 SF
3. Wooded area	NOT ESTABLISHED
4. Floodplain area	NOT ESTABLISHED
5. Net lot area	20,274 SF
6. TOTAL AREA DISTURBED	10,652 SF (AT LEAST)
7. SWM REVIEW	CONFIRM WITH LOUDOUN COUNTY

BUILDING DATA

	G	B	FF	SF	HT	STY
42525 CARNFORTH	—	—	—	—	—	

CT ASHBURN VA
20148

DRAINAGE AREA COMPUTATIONS

TREATMENT	SF	ACRES	C
Total — 28.4% impervious	20,274	0.465	
PRE-DEVELOPMENT	SF		
Pervious — lawn / meadow	20,274		0.25
Composite C			0.250
POST-DEVELOPMENT	SF		
Proposed roof	5,069		0.06
Driveway and on-lot paving	698		0.005
Pervious — lawn	14,508		0.25
Composite C			0.449
Flow path		235 ft	10.2 %
Flow path slope			10.2 %
Time of concentration (Kiprich)			5.0 min
Rainfall intensity, 10-yr			6.53 in/hr
Peak discharge PRE, Q = CIA			0.75 cfs
Peak discharge POST, Q = CIA			1.38 cfs
INCREASE TO BE MANAGED			0.66 cfs
Rv = 0.25 ± 0.051			0.260
Water quality volume WqV			571 cf

Time of concentration by Kirpich over a 235 ft flow path at 10.2% slope, taken from the 2 contours across the catchment. A 5-minute flow apes.
 Rainfall intensity 6.53 in/hr for the 10-year storm at a 5.0 minute duration. NOAA Atlas 14 Vol. 2 Ver. 3, PFDS point estimate, 30.0388 N 75.7332 W.
 Water quality volume on a 1.0 in rainfall. Computed by the Maryland Stormwater Design Manual method for reference only; LOUDOUN COUNTY's own stormwater method governs this lot.
 Runoff coefficients: roof and pavement 0.95, average lawn 0.25. Confirm the lawn coefficient against the soil hydrologic group before relying on the composite.
 Sidewalk, curb and gutter and the driveway appear to lie in the public right of way and drain to the street. They are excluded from the lot catchment and included in the disturbed area.

SEQUENCE OF CONSTRUCTION

1. Pre-construction meeting
2. Obtain necessary permits
3. Notify Miss Utility at 811 at least 48 hours prior to any excavation
4. Install stabilized construction entrance and perimeter sediment control
5. Rough grade the lot
6. Construct dwelling
7. Install water, sewer and storm services and the driveway
8. Fine grade and stabilize all disturbed areas
9. Remove sediment control once the inspector gives written permission

TOTAL ESTIMATED TIME OF CONSTRUCTION: 8 MONTHS

GENERAL NOTES

1. CONTRACTOR SHALL CONTACT UTILITY AT 811 A MINIMUM OF 72 HOURS PRIOR TO CONSTRUCTION. FIELD VERIFY LOCATION AND DEPTH BY TEST PIT BEFORE CONSTRUCTION.
2. EXISTING UTILITIES ARE SHOWN FROM PUBLIC RECORD AND ARE APPROXIMATE. LOCATION AND DEPTH SHALL BE FIELD-VERIFIED BY CONTRACTOR.
3. PROPOSED GRADE SHOWN SOLD; EXISTING GRADE SHOWN DASHED.
4. 18" HIGH EARTHWORK GRADES AND UTILITY ELEVATIONS SHOWN TO TENTHS OF A FOOT.
5. BOUNDARY SHOWN IS COMPLIED JURISDICTION GIS, NOT A BOUNDARY SURVEY.
6. CONCRETE CURB, FILING PAVEMENT, CURB AND GUTTER, DRIVEWAY, AND SIDEWALK IN LINE AND GRADE.
7. NO DRIVEWAY CULVERT IS REQUIRED. THIS FRONTAGE IS AN EXISTING CURB AND GUTTER SECTION, NOT AN OPEN DITCH SECTION. GUTTER SHALL BE 12" DEEP AND 12" WIDE. THE EXISTING DEPRESSION CURB AT THE APRON PER LOUDOUN COUNTY STANDARD, SHALL BE MAINTAINED. THE EXISTING DRIVEWAY SHALL BE MAINTAINED. REAR DRAINAGE EASEMENT AND CROSSES NO DRIVEWAY OR APRON.

PROFESSIONAL CERTIFICATION

I HEREBY CERTIFY THAT THESE DOCUMENTS WERE PREPARED OR APPROVED BY ME, AND THAT I AM A DULY LICENSED PROFESSIONAL ENGINEER UNDER THE LAWS OF THE COMMONWEALTH OF VIRGINIA.

SEAL	SIGNATURE _____	DATE _____
	LICENSE NO. _____	EXPIRATION DATE _____

REVISIONS

— none —



SOILS TABLE USDA NRCS SSURGO

MAP UNIT	MAP UNIT NAME	K-FACT	HYDRIC	HYD. GRP	DRAINAGE CLASS
101	Pits, quarry	—	No	—	—
102	Landfills	—	No	—	—
103	Udorthens	—	No	—	—
106	Mongie silt loam, 0 to 7 percent slopes, frequently flooded	32	No	B/D	Somewhat poorly drained
132	Rotaryville cobbly silt loam, 0 to 7 percent slopes	43	No	C/D	Somewhat poorly drained
138	Rotaryville cobbly silt loam, 7 percent slopes	37	No	B	Well drained
148	Manassas silt loam, 2 to 7 percent slopes	37	No	B	Moderately well drained
17B	Middleburg silt loam, 2 to 7 percent slopes	32	No	A	Well drained

SOURCE AND ACCURACY
Parcel boundary — jurisdiction gis — Loudoun County — parcel boundaries — AVONMORE, Plat 2023-0020, L...
Existing contours — U.S. Geological Survey — 3D Elevation Program — 3DEP bare-earth DEM —...

GRAPHIC SCALE — 1" = 30'





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design, build, deliver

PROJECT
42525 CARNFORTH CT, ASHBURN, VA, 20148

42525 CARNFORTH CT, ASHBURN, VA, 20148

JURISDICTION
Loudoun County, Virginia

ZONE
R1

SHEET
C-500 — Utility Plan

SCALE
1" = 30'

STATUS
PRELIM

SHEET NO.

COORDINATING EDITOR

EPSG:2248
H: NAD83

V: NAVD88 (feet)

Virginia Prof

certifies: utility_design

SEAL AND SIGNATURE

All content on this sheet falls within one professional scope

PLAN TYPE AND PREPARATION

PLAN TYPE	SITE DEVELOPMENT / FINE GRADING
SHEET	C-500 — Utility Plan
DISCIPLINE	Virginia Professional Engineer
DRAWN BY	KEALEE SITE-PLAN ENGINE
DESIGNED BY	
CHECKED BY	

REVISIONS

[illegible]

SITE DATA

ZONE	R1	
TAX / PARCEL ID	154363671000	
GROSS TRACT AREA	20,274 SF	0.465 AC
LOT FRONTAGE	79'	
SETBACKS	REQUIRED	PROVIDED
FRONT YARD	35'	35'
SIDE YARD	12'	12'
REAR YARD	35'	35'
COVERAGE	MAX	PROPOSED
LOT COVERAGE	25%	25.0%
BUILDING FOOTPRINT		5,069 SF
DATUM		
HORIZONTAL	EP5G-2248	NAD83
VERTICAL	NAVD88 (feet)	

SITE ANALYSIS

1. Gross tract area	20,274 SF
2. Area of dwelling	5,069 SF
3. Wooded area	NOT ESTABLISHED
4. Floodplain area	NOT ESTABLISHED
5. Net lot area	20,274 SF
6. TOTAL AREA DISTURBED	10,652 SF (AT LEAST)
7. SWM REVIEW	CONFIRM WITH LOUDOUN COUNTY

BUILDING DATA

	G	B	FF	SF	HT	STY
42525 CARNFORTH	—	—	—	—	—	—
CT, ASHBURN, VA	—					
20148	—					

— = Measurement - FF finished floor - SF subfloor - HT height - STY storeys. A DASH IS NOT ZERO: the elevation has not been established and must be set from a field-run survey before construction.

DRAINAGE AREA COMPUTATIONS

CATCHMENT	SF	ACRES	C
Total ~ 28.4% impervious	20,274	0.465	
PRE-DEVELOPMENT	SF		
Pervious ~ lawn / meadow	20,274		0.250
Composite C			0.250
POST-DEVELOPMENT	SF		
Proposed roof	5,069		0.95
Driveway and on-lot paving	698		0.16
Pervious ~ lawn	14,508		0.25
Composite C			0.449
Flow path			25 ft
Flow path slope			10 ± %
Time of concentration (Kirpich)			5.0 min
Rainfall intensity, 10-yr			6.33 in/hr
Peak discharge PRE, Q = CIA			0.75 cfs
Peak discharge POST, Q = CIA			1.38 cfs
INCREASE TO BE MANAGED			0.66 cfs
Rv = 0.05 ~ 0.0501			0.3500
Water quality volume WQv			517 cft

Time of concentration by Kirpich over a 235 ft flow path at 10.2% tan, taken from the 2 ft contours across the catchment. A 5-minute flow path. Rainfall intensity 6.53 in/hr for the 10-year storm at a 5.0 minute duration. NOAA Atlas 14 Vol. 2 Ver. 3, PFDs point estimate, 93.0386 N 75.7352 W. Water quality volume on a 1.0 in rainfall. Computed by the Maryland Stormwater Design Manual method for reference only; LOUDOUN COUNTY's own stormwater method gives this lot.

Runoff coefficients: roof and pavement 0.95, average lawn 0.25. Confirmed the lawn coefficient against the soil hydrologic group before relying on the composite. Sidewalk, curb and gutter and the driveway apron lie in the public right-of-way and drain to the street. They are excluded from the lot catchment and included in the disturbed area.

SEQUENCE OF CONSTRUCTION

1. Pre-construction meeting
2. Obtain necessary permits
3. Notify Miss Utility at 811 at least 48 hours prior to any excavation
4. Install stabilized construction entrance and perimeter sediment control
5. Rough grade the lot
6. Construct dwelling
7. Install water, sewer and storm services and the driveway
8. Fine grade and stabilize all disturbed areas
9. Remove sediment control once the inspector gives written permission

TOTAL ESTIMATED TIME OF CONSTRUCTION: 8 MONTHS

GENERAL NOTES

1. CONTRACTOR SHALL CONTACT MISS UTILITY AT 811 A MINIMUM OF 48 HOURS PRIOR TO ANY EXCAVATION; FIELD-VERIFIED LOCATION AND DEPTH SHALL BE SHOWN.
2. EXISTING UTILITIES ARE SHOWN FROM PUBLIC RECORD AND ARE APPROXIMATE. LOCATION AND DEPTH SHALL BE FIELD-VERIFIED BY THE CONTRACTOR.
3. PROPOSED GRADE SHOWN SOLID; EXISTING GRADE SHOWN DASHED.
4. ROUGH EARTHWORK GRADES AND UTILITY ELEVATIONS SHOWN TO TENTH OF A FOOT.
5. BOUNDARY SHOWN IS COMPLIED JURISDICTION GIS, NOT A BOUNDARY SURVEY.
6. COVERED EXISTING PAVEMENT, CURBS AND GUTTER, DRIVEWAY, AND SIDEWALK IN LINE AND GRADE.
7. NO DRIVEWAY CULVERT IS REQUIRED; THIS FRONTAGE IS AN EXISTING CURB AND GUTTER SECTION; NOT AN OPEN DITCH SECTION. DRIVEWAY SHALL BE CARRIED THROUGH CHUTE ENTRANCE BY THE DEPRESSION CURB AT THE APRON PER LOCAL COUNTY STANDARD. DRIVEWAY SHALL BE CARRIED THROUGH THE REAR DRAINAGE APRON, REAR DRIVEWAY EASEMENT AND DRIVEWAY TO DRIVEWAY OR APRON.

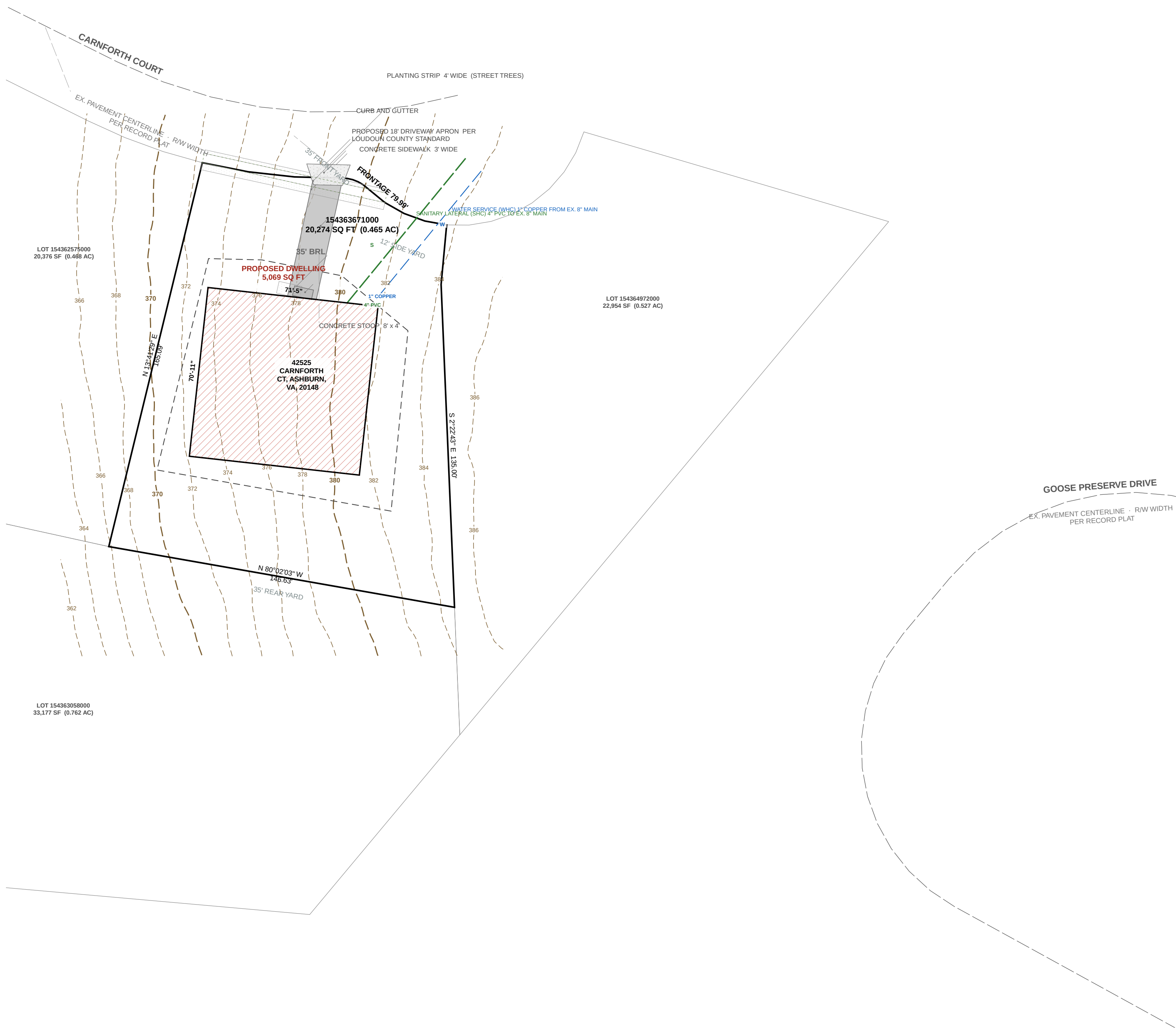
PROFESSIONAL CERTIFICATION

I HEREBY CERTIFY THAT THESE DOCUMENTS WERE PREPARED OR APPROVED BY ME, AND THAT I AM A DULY LICENSED PROFESSIONAL ENGINEER UNDER THE LAWS OF THE COMMONWEALTH OF VIRGINIA.

SEAL	SIGNATURE _____	DATE _____
	LICENSE NO. _____	EXPIRATION DATE _____

REVISIONS

— none —



SOILS TABLE USDA NRCS SSURGO

MAP UNIT	MAP UNIT NAME	K-FACT	HYDRIC	HYD. GRP	DRAINAGE CLASS
101	Pits, quarry	—	No	—	—
102	Landfills	—	No	—	—
103	Udorthens	—	No	—	—
106	Mongie silt loam, 0 to 7 percent slopes, frequently flooded	32	No	B/D	Somewhat poorly drained
132	Rothville cobbly silt loam, 0 to 7 percent slopes	43	No	C/D	Somewhat poorly drained
138	Richwood silt loam, 7 to 15 percent slopes	37	No	B	Well drained
148	Manassas silt loam, 2 to 7 percent slopes	37	No	B	Moderately well drained
17B	Middleburg silt loam, 2 to 7 percent slopes	32	No	A	Well drained

SOURCE AND ACCURACY
Parcel boundary — jurisdiction gis — Loudoun County — parcel boundaries — AVONMORE, Plat 2023-0020, L...
Existing contours — U.S. Geological Survey — 3D Elevation Program — 3DEP bare-earth DEM —...

GRAPHIC SCALE — 1" = 30'



- 1 DAY
2 DAYS
1 DAY
2 DAYS
5 DAYS
MONTHS
10 DAYS
5 DAYS
1 DAY